

Lab 2: LINEAR REGRESSION

DUE: 13:00h 17/04/2026

Submission: mlearning

- **Lab2_YOURNAME.ipynb**
- Must include all data file (.csv,...) and .ipynb in your submission
- .csv and .ipynb must be in the same folder (pd.read_csv('data.csv'))
- No use of LOCAL libraries/folders
- NO LATE submission accepted

Apply Linear Regression to predict house prices for each of the following cases.

For each case:

1. normalize data using MinMaxScaler()
2. use 80% data for training and 20% for testing
3. write the **Linear Equation** for predicting house prices
4. compute **MSE** score for testing data
5. compute the **coefficient of determination R2 score** for testing data for each case

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

There are 6 cases:

1. Use only area
2. Use only area, bedrooms
3. Use only area, bedrooms, bathrooms
4. Use only area, bedrooms, bathrooms, stories
5. Use only area, bedrooms, bathrooms, stories, parking
6. Use only area, bedrooms, bathrooms, stories, parking, airconditioning (yes->1, no->0)

Summarize MSE, R2 metrics of 6 cases in a Table